

CCNA 200-301 V2.0

16 Static Routing, IPv4 and IPv6

Part III — IP Routing

EXAM TOPICS

3.2

LECTURER

Dr. Mustafa Hameed

THE ISLAMIA UNIVERSITY OF BAHAWALPUR

Department of Information Technology

What you will be able to do

- **Configure** the four kinds of static route the blueprint names: default, network, host and floating.
- **Choose** correctly between a next-hop address, an exit interface, and both — a choice that matters more than it looks.
- **Configure** static routing for IPv6 alongside IPv4.
- **Troubleshoot** a static route that is absent from the table, or present and not working.

Before we start

Chapter 15 — longest prefix match and administrative distance are assumed throughout.
Chapter 5 for the v6 half.

Vocabulary for this chapter

static route

default route

network route

host route

floating static

next hop

exit interface

fully specified route

recursive lookup

gateway of last resort

Each is defined where it first matters — not here.

The verb is troubleshoot, and the sub-topics are a checklist

This is what the blueprint actually asks for

Topic **3.2** is “troubleshoot IPv4 and IPv6 static routing” and then lists exactly four kinds: default route, network route, host route, floating static. Expect to be shown a table and a symptom rather than asked to recite syntax — but you cannot diagnose what you cannot write, so learn the syntax first.

All four, on ipv4

Configure

```
! Network route -- reach 10.9.9.0/24 through the neighbour at 10.255.0.2.
ip route 10.9.9.0 255.255.255.0 10.255.0.2
!
! Default route -- the gateway of last resort.
ip route 0.0.0.0 0.0.0.0 203.0.113.1
!
! Host route -- this one server goes a different way.
ip route 10.9.9.50 255.255.255.255 10.255.1.2
!
! Floating static -- distance 200 keeps it out of the table while
! OSPF (distance 110) has a route. It appears only if OSPF loses one.
ip route 10.1.0.0 255.255.0.0 10.255.2.2 200
```

Why the host route wins

All four routes above can match `10.9.9.50`. The `/32` host route is the longest prefix, so it is used — ahead of the `/24`, ahead of the default. Longest prefix match decides before distance is even considered (Chapter 15).

Exit-interface-only routes on Ethernet are a trap

Ethernet is multi-access: many devices share the segment. A route that names only an exit interface tells the router “send it out here” without saying to whom, so the router must ARP for *every destination address* it forwards, hoping something answers. On a link where proxy ARP is enabled it appears to work; where it is not, it silently fails; and either way the ARP table fills with entries that should not exist.

On a serial point-to-point link there is only one possible recipient, so the form is harmless. On Ethernet, always give the next hop — alone or fully specified.

show ip route static

```
R1# show ip route static
```

```
      10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
S       10.1.0.0/16 [200/0] via 10.255.2.2
S       10.9.9.0/24 [1/0] via 10.255.0.2
S       10.9.9.50/32 [1/0] via 10.255.1.2
S*     0.0.0.0/0 [1/0] via 203.0.113.1
```

Ipv6 static routes

Configure

```
ipv6 unicast-routing
!  
! Network route via a global next-hop address.  
ipv6 route 2001:db8:acad:20::/64 2001:db8:acad:99::2  
!  
! Default route.  
ipv6 route ::/0 2001:db8:acad:99::2  
!  
! Host route.  
ipv6 route 2001:db8:acad:20::50/128 2001:db8:acad:99::2  
!  
! Floating static.  
ipv6 route 2001:db8:acad:30::/64 2001:db8:acad:98::2 200  
!  
! Via a LINK-LOCAL next hop -- the exit interface is mandatory,  
! because fe80:: addresses are only unique per link and the router  
! has no way to know which interface you mean.  
ipv6 route 2001:db8:acad:40::/64 GigabitEthernet0/1 fe80::2
```

Why link-local next hops need the interface

A link-local address is valid on one link only, and the same `fe80::2` may legitimately exist on three different interfaces of the same router. Naming the interface removes the ambiguity. IOS rejects a link-local next hop without one.

This matters because routing protocols, including OSPFv3, use link-local addresses as their next hop — so link-local next hops are the normal case in IPv6, not an oddity.

show ipv6 route static

```
R1# show ipv6 route static
```

```
IPv6 Routing Table - default - 7 entries
```

```
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
```

```
S    ::/0 [1/0]
```

```
    via 2001:DB8:ACAD:99::2
```

```
S    2001:DB8:ACAD:20::/64 [1/0]
```

```
    via 2001:DB8:ACAD:99::2
```

```
S    2001:DB8:ACAD:40::/64 [1/0]
```

```
    via FE80::2, GigabitEthernet0/1
```

A backup path that only activates on failure

R1 reaches `10.1.0.0/16` through OSPF over its primary link, and has a slow secondary link to the same destination.

- OSPF's route has distance 110.
- Configure the backup with a distance *above* 110: `ip route 10.1.0.0 255.255.0.0 10.255.2.2 200`.
- While OSPF is up, only the OSPF route is installed. `show ip route` shows `0 10.1.0.0/16 [110/20]` and no static entry at all.
- The primary link fails. OSPF withdraws its route. The static route, now the only candidate, is installed as `S 10.1.0.0/16 [200/0]`.

A backup path that only activates on failure (continued)

- The primary returns. OSPF's route at 110 beats 200 and displaces the static again.

The number 200 is not magic — it just has to exceed the distance of whatever it is backing up. Back up a *static* route (distance 1) and any value from 2 upwards will do.

Floating statics, properly

- **A floating static with too low a distance.** Set it below the dynamic protocol's and it takes over permanently, which is the opposite of a backup.
- **Exit-interface-only routes on Ethernet.** See the warning above.
- **Forgetting the reverse route.** A static route is one-directional. Both ends need one, and forgetting this is the commonest static-routing fault of all.
- **A next hop that is not directly reachable.** The route will not be installed at all; the router silently discards it. `show ip route` simply has no entry, which looks like you never typed it.
- **Omitting the interface on an IPv6 link-local next hop.** Rejected outright.

The static route was typed and it is not in the table.

An engineer configures `ip route 10.9.9.0 255.255.255.0 10.255.0.2` and `show ip route static` shows nothing. No error was displayed when the command was entered.

The static route was typed and it is not in the table.

What would you check first — and what do you expect to see?

show ip route 10.255.0.2

```
R1# show ip route 10.255.0.2
```

```
% Network not in table
```

show ip interface brief | include Gigabit

```
R1# show ip interface brief | include Gigabit
```

```
GigabitEthernet0/1    10.255.0.1    YES manual administratively down down
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. A static route is only installed if the router can resolve its next hop. `10.255.0.2` is not reachable — there is no connected subnet covering it — so the route is accepted into the configuration and never enters the routing table. This is why the command produced no error: the syntax was perfect and the topology was not.

Check the interface that should reach it:

And the lesson

Fix. The interface is shut. `no shutdown` on `Gi0/1`, and the connected route for `10.255.0.0/30` appears, the next hop resolves, and the static route installs itself immediately. Confirm with `show ip route static` — and remember the general rule: **a static route missing from the table is nearly always an unreachable next hop**, not a typing error.

Four routes, one failure

Topology. Three routers — R1, R2, R3 — with R1 connected to R2 by a fast link and to R3 by a slow one, and R2 and R3 both connected to a LAN at the far side. A LAN on R1.

1. Address everything. Confirm connected routes on all three and that neighbours ping.
2. On R1, add a network route to the far LAN via R2. Test from the R1 LAN — it will fail. Diagnose why, then add the reverse route and re-test.
3. Add a default route on R2 and R3 pointing back at R1. Note the S* and the gateway of last resort line.
4. Add a host route on R1 for one server in the far LAN, via R3. Prove with `show ip route <address>` that it is preferred over the network route.
5. Add a floating static via R3 for the far LAN with distance 5. Confirm it is *not* in the table.

Four routes, one failure (continued)

1. Shut **R1**'s link to **R2**. Confirm the floating static appears and traffic still flows. Restore the link and watch it withdraw.
2. **Extension.** Repeat steps 2 and 5 for IPv6, including one route via a link-local next hop.

CHECKPOINT 1 of 3

You configure a static route and it never appears in the table, with no error message. What is the most likely cause and which command confirms it?

Discuss · then advance

Checkpoint 1 of 3

You configure a static route and it never appears in the table, with no error message. What is the most likely cause and which command confirms it?

The next hop is not reachable. A router will not install a static route whose next-hop address it cannot resolve to an interface. Confirm with `show ip route <next-hop>` — if that returns nothing, the static route has nowhere to point.

CHECKPOINT 2 of 3

A floating static backing up an OSPF route is configured with distance 100. What will happen, and why is it wrong?

Discuss · then advance

Checkpoint 2 of 3

A floating static backing up an OSPF route is configured with distance 100. What will happen, and why is it wrong?

Nothing will happen: 100 is *lower* than OSPF's 110, so the "backup" route wins permanently and the OSPF route is never installed. A floating static must have a distance *higher* than the protocol it backs up.

CHECKPOINT 3 of 3

Why must an IPv6 static route using a link-local next hop also name an exit interface?

Discuss · then advance

Checkpoint 3 of 3

Why must an IPv6 static route using a link-local next hop also name an exit interface?

Because a link-local address is only unique within a link, and a router with several interfaces has no way to tell which one `fe80::2` refers to. Naming the exit interface removes the ambiguity.

What to take away

- Four kinds: network, default (`0.0.0.0/0`), host (`/32`) and floating. The host route wins on longest prefix match.
- Give a next hop on Ethernet — alone or fully specified. Exit-interface only belongs on point-to-point links.
- IPv6 mirrors IPv4, but a link-local next hop *must* name the exit interface.
- A floating static needs a distance above whatever it backs up: over 110 for OSPF, over 1 for another static.
- A static route absent from the table almost always means an unresolvable next hop. Routes are one-way; configure both ends.

Where we got to

- Four kinds: network, default (0.0.0.0/0), host (/32) and floating. The host route wins on longest prefix match.
- Give a next hop on Ethernet — alone or fully specified. Exit-interface only belongs on point-to-point links.
- IPv6 mirrors IPv4, but a link-local next hop *must* name the exit interface.
- A floating static needs a distance above whatever it backs up: over 110 for OSPF, over 1 for another static.

NEXT

Chapter 17 — OSPFv2, Single Area